```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
#Excel dataset Upload

df=pd.read_excel("/content/sales_data.xlsx")
```

```
#Data Cleaning

df.head()
```

	Order_ID	Order_Date	Customer_Name	Category	Product	Region	Quantity	Sales	Profit
0	ORD001	2025-01-05	Rahul Sharma	Technology	Laptop	West	2	80000	12000
1	ORD00	2025-02-12	Priya Patil	Furniture	Chair	South	5	25000	5000
2	ORD003	2025-02-20	Suvarna Ukirde	Office Supplies	Notebook	East	10	5000	1500
3	ORD004	2025-02-20	Tanuja Ukirde	Technology	Mobile	North	3	45000	9000
4	ORD005	2025-03-22	Rupali Vitnor	Furniture	Table	West	2	35000	6000

```
df.shape
```

```
(10, 9)
```

```
df.columns
```

```
Index(['Order_ID', 'Order_Date', 'Customer_Name', 'Category', 'Product',
       'Region', 'Quantity', 'Sales', 'Profit'],
      dtype='object')
```

```
#Basic Analysis
df.isnull()
```

	Order_ID	Order_Date	Customer_Name	Category	Product	Region	Quantity	Sales	Profit
0	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False
5	False	False	False	False	False	False	False	False	False
6	False	False	False	False	False	False	False	False	False
7	False	False	False	False	False	False	False	False	False
8	False	False	False	False	False	False	False	False	False
9	False	False	False	False	False	False	False	False	False

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Order_ID       10 non-null     object
 1   Order_Date     10 non-null     datetime64[ns]
 2   Customer_Name  10 non-null     object
 3   Category       10 non-null     object
 4   Product        10 non-null     object
 5   Region         10 non-null     object
 6   Quantity       10 non-null     int64
 7   Sales          10 non-null     int64
 8   Profit         10 non-null     int64
```

```
dtypes: datetime64[ns](1), int64(3), object(5)
memory usage: 852.0+ bytes
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df.describe()
```

	Order_Date	Quantity	Sales	Profit
count	10	10.000000	10.000000	10.000000
mean	2025-03-21 14:24:00	5.400000	29700.000000	5350.000000
min	2025-01-05 00:00:00	1.000000	2000.000000	500.000000
25%	2025-02-20 00:00:00	2.000000	13750.000000	2875.000000
50%	2025-03-30 00:00:00	3.500000	27500.000000	5500.000000
75%	2025-04-16 00:00:00	5.000000	38750.000000	6750.000000
max	2025-05-25 00:00:00	20.000000	80000.000000	12000.000000
std	NaN	5.738757	22871.865298	3488.473082

```
#KPL Cards

total_sales = df['Sales'].sum()
total_profit = df['Profit'].sum()
total_orders = df['Order_ID'].nunique()
avg_order_value = df['Sales'].mean()

print("Total_Sales:",total_sales)
print("Total_Profit:",total_profit)
print("Total_Orders:",total_orders)
print("Avg_order_Values:",avg_order_value)
```

```
Total_Sales: 297000
Total_Profit: 53500
Total_Orders: 10
Avg_order_Values: 29700.0
```

```
# interactive filters
categories = df['Category'].unique()

print("Available Categories:")
print(categories)
```

```
Available Categories:
['Technology' 'Furniture' 'Office Supplies']
```

```
regions = df['Region'].unique()

print("Available Regions:")
print(regions)
```

```
Available Regions:
['West' 'South' 'East' 'North']
```

```
!pip install ipywidgets
```

Show hidden output

```
import ipywidgets as widgets
from IPython.display import display
```

```
category_filter = widgets.Dropdown(
    options=["All"] + list(df["Category"].unique()),
    description="Category:"

)
```

```
region_filter = widgets.Dropdown(
    options=["All"] + list(df["Region"].unique()),
    description="Region:"

)
display(category_filter)
display(region_filter)
```

Category: All

Region: All

```
def filter_data(category, region):
  filtered_df = df.copy()
  #apply category filter
  if category !="All":
    filtered_df = filtered_df[filtered_df["Category"]==category]
    #apply region filter
    if region_filter !="All":
      filtered_df = filtered_df[filtered_df["Region"]==region]

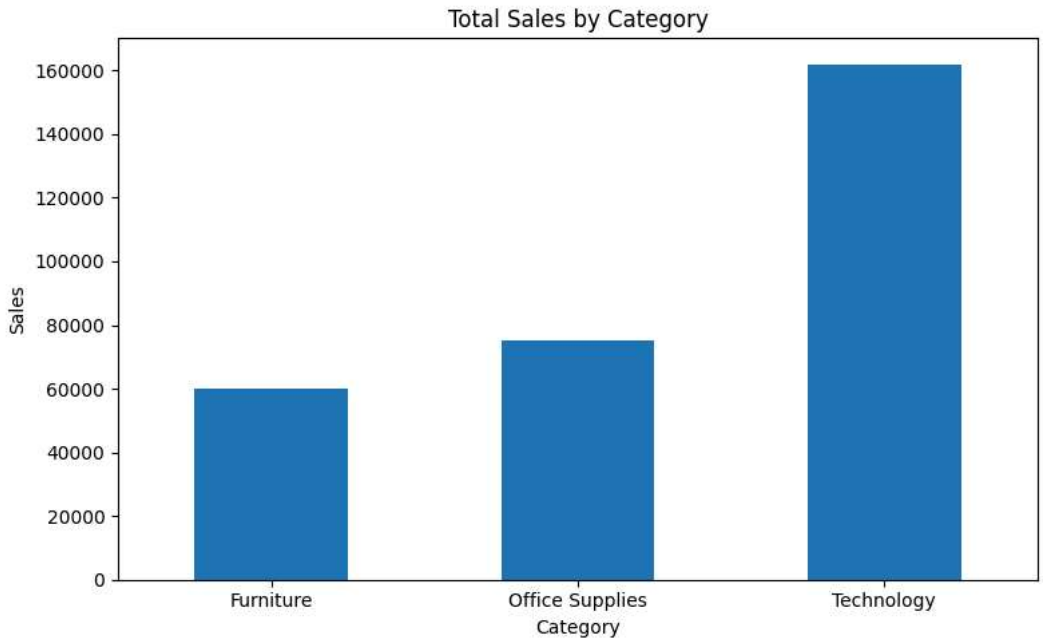
      return filtered_df
```

```
filtered_df = filter_data(
    category_filter.value,
    region_filter.value
)
filtered_df
```

```
from pandas.core.indexes import category
category_sales = df.groupby("Category")['Sales'].sum()

plt.figure(figsize=(8,5))
category_sales.plot(kind="bar")

plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



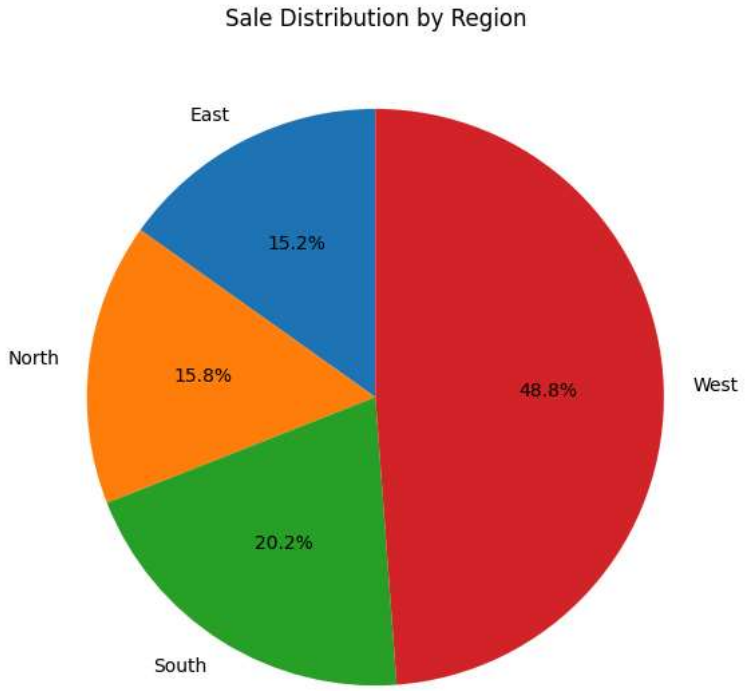

```
region_sales = df.groupby("Region")['Sales'].sum()
print(region_sales)

region_sales = df.groupby("Region")['Sales'].sum()
```

```
plt.figure(figsize=(7,7))

plt.pie(
    region_sales,
    labels=region_sales.index,
    autopct="%1.1f%%",
    startangle=90
)
plt.title("Sale Distribution by Region")
plt.show()
```

```
Region
East      45000
North     47000
South     60000
West     145000
Name: Sales, dtype: int64
```



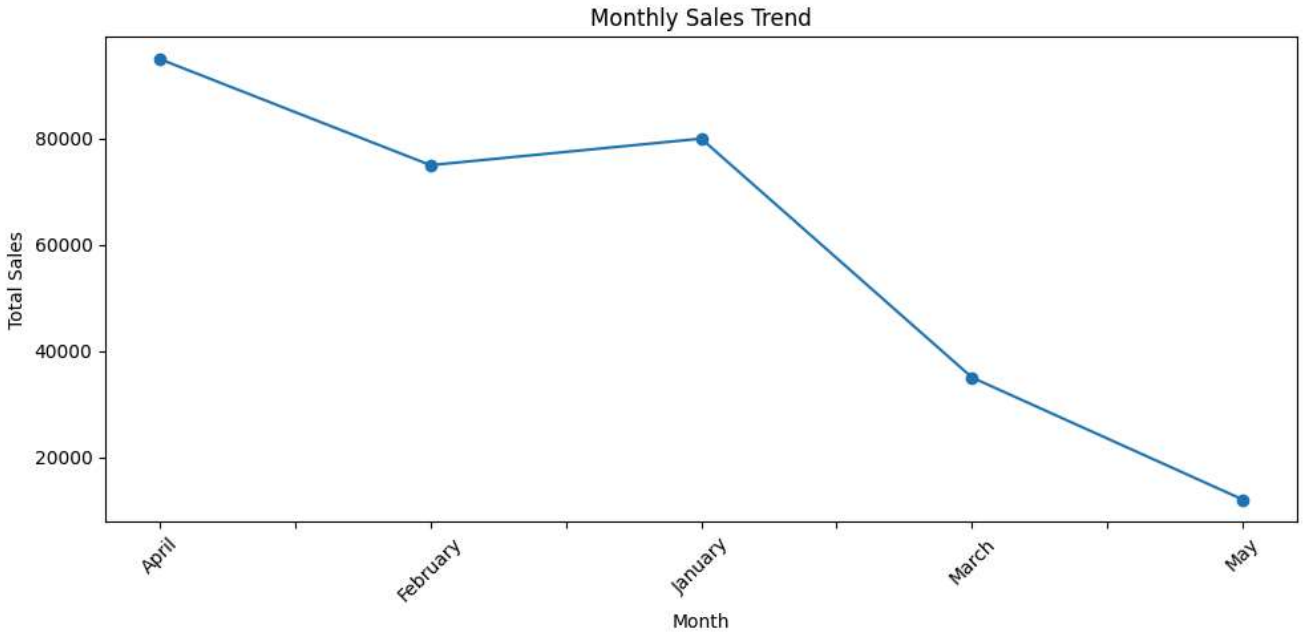

```
df["Order_Date"] = pd.to_datetime(df["Order_Date"])

df["Month"]= df["Order_Date"].dt.strftime("%B")
df[["Order_Date","Month","Sales"]]
monthly_sales = df.groupby("Month")["Sales"].sum()
print(monthly_sales)

plt.figure(figsize=(10,5))
monthly_sales.plot(kind="line",marker="o")

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
Month
April       95000
February    75000
January     80000
March       35000
May         12000
Name: Sales, dtype: int64
```



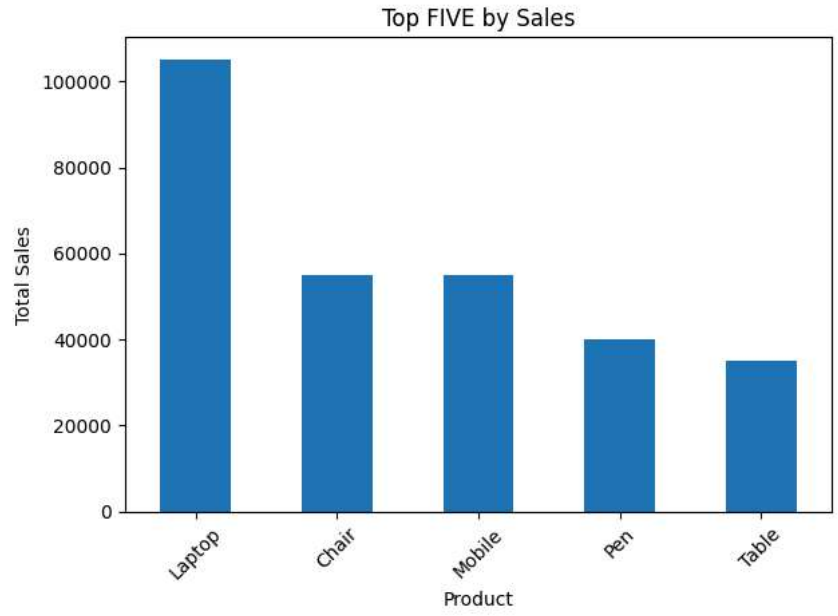

```
product_sales = df.groupby("Product")["Sales"].sum().sort_values(ascending=False)
print(product_sales)
```

```
Product
Laptop        105000
Chair          55000
Mobile         55000
Pen            40000
Table          35000
Notebook        5000
Headphones      2000
Name: Sales, dtype: int64
```

```
top_5_products = product_sales.head(5)
print(top_5_products)
```

```
Product
Laptop    105000
Chair      55000
Mobile     55000
Pen        40000
Table      35000
Name: Sales, dtype: int64
```

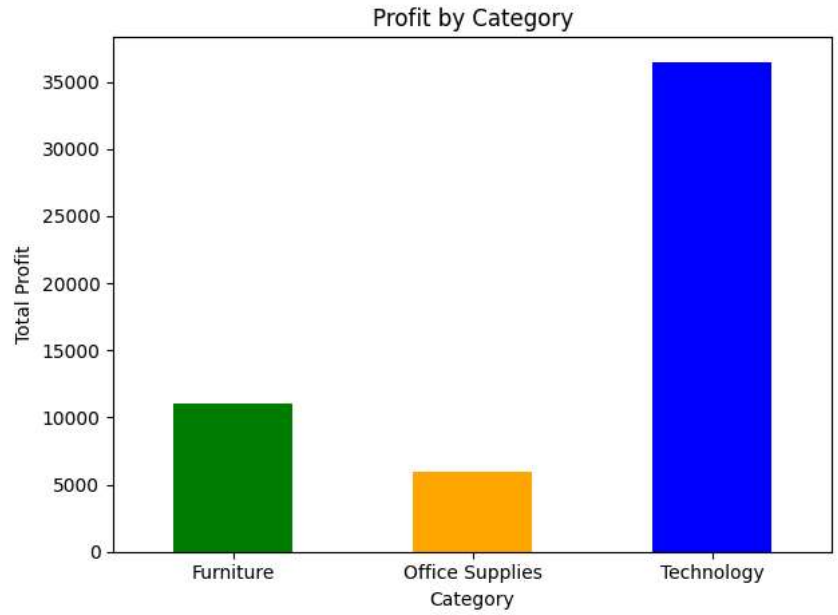
```
top_5_products.plot(kind="bar")
plt.title("Top FIVE by Sales")
plt.xlabel("Product")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```




```
profit_by_category = df.groupby("Category")['Profit'].sum()
print(profit_by_category)
```

```
Category
Furniture          11000
Office Supplies     6000
Technology         36500
Name: Profit, dtype: int64
```

```
profit_by_category.plot(kind="bar",color=["green","orange","blue"])
plt.title("Profit by Category")
plt.xlabel("Category")
plt.ylabel("Total Profit")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```




```
total_profit = df["Profit"].sum()

profit_percentage = (df.groupby("Category")["Profit"].sum()/total_profit)*100

print(profit_percentage.round(2))
```

```
Category
Furniture          20.56
Office Supplies    11.21
Technology         68.22
Name: Profit, dtype: float64
```

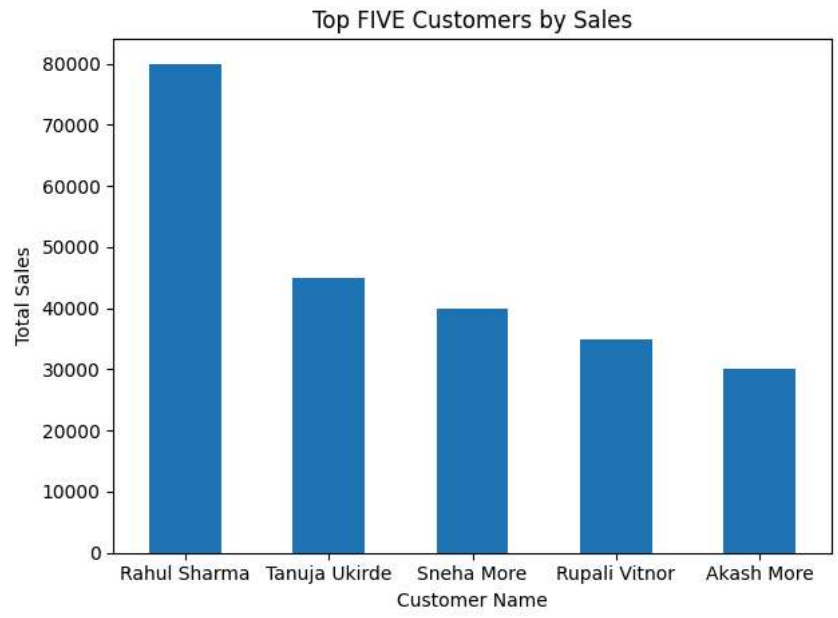
```
customer_sales=(
    df.groupby("Customer_Name")["Sales"].sum().sort_values(ascending=False)
)
print(customer_sales)
```

```
Customer_Name
Rahul Sharma      80000
Tanuja Ukirde     45000
Sneha More        40000
Rupali Vitnor     35000
Akash More        30000
Amit Joshi        25000
Priya Patil       25000
Rohan Patil       10000
Suvarna Ukirde     5000
Pooja Kale         2000
Name: Sales, dtype: int64
```

```
top_5_customers=customer_sales.head(5)
print(top_5_customers)
```

```
Customer_Name
Rahul Sharma     80000
Tanuja Ukirde    45000
Sneha More       40000
Rupali Vitnor    35000
Akash More       30000
Name: Sales, dtype: int64
```

```
top_5_customers.plot(kind="bar")
plt.title("Top FIVE Customers by Sales")
plt.xlabel("Customer Name")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```




```
def update_dashboard(category, region):

    # Filter data
    filtered_df = df.copy()

    if category != "All":
        filtered_df = filtered_df[
            filtered_df["Category"] == category
```

```
        ]

    if region != "All":
        filtered_df = filtered_df[
            filtered_df["Region"] == region
        ]

    # KPI calculations
    total_sales = filtered_df["Sales"].sum()
    total_profit = filtered_df["Profit"].sum()
    total_orders = filtered_df["Order_ID"].nunique()

    if total_orders > 0:
        avg_order = total_sales / total_orders
    else:
        avg_order = 0

    # Display KPI
    print("=" * 60)
    print("             SALES ANALYSIS DASHBOARD")
    print("=" * 60)

    print("Category :", category)
    print("Region   :", region)
    print()

    print("💰 Total Sales         : ₹", f"{total_sales:,.0f}")
    print("📈 Total Profit        : ₹", f"{total_profit:,.0f}")
    print("📦 Total Orders        :", total_orders)
    print("🛒 Average Order Value : ₹", f"{avg_order:,.0f}")

    print("=" * 60)

    # Sales by Category
    category_sales = filtered_df.groupby(
        "Category"
    )["Sales"].sum()

    plt.figure(figsize=(8, 5))
    category_sales.plot(kind="bar")
    plt.title("Sales by Category")
    plt.xlabel("Category")
    plt.ylabel("Sales (₹)")
    plt.xticks(rotation=0)
    plt.tight_layout()
    plt.show()

    # Sales by Region
    region_sales = filtered_df.groupby(
        "Region"
    )["Sales"].sum()

    plt.figure(figsize=(7, 7))
    plt.pie(
        region_sales,
        labels=region_sales.index,
        autopct="%1.1f%%"
    )
    plt.title("Sales by Region")
    plt.show()

    # Top Products
    product_sales = filtered_df.groupby(
        "Product"
    )["Sales"].sum().sort_values(ascending=False).head(5)

    plt.figure(figsize=(8, 5))
    product_sales.plot(kind="bar")
    plt.title("Top 5 Products")
    plt.xlabel("Product")
    plt.ylabel("Sales (₹)")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()

    # Top Customers
    customer_sales = filtered_df.groupby(
```

```
        "Customer_Name"
    )["Sales"].sum().sort_values(ascending=False).head(5)

    plt.figure(figsize=(8, 5))
    customer_sales.plot(kind="bar")
    plt.title("Top 5 Customers")
    plt.xlabel("Customer")
    plt.ylabel("Sales (₹)")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()
```

```
import ipywidgets as widgets
from IPython.display import display

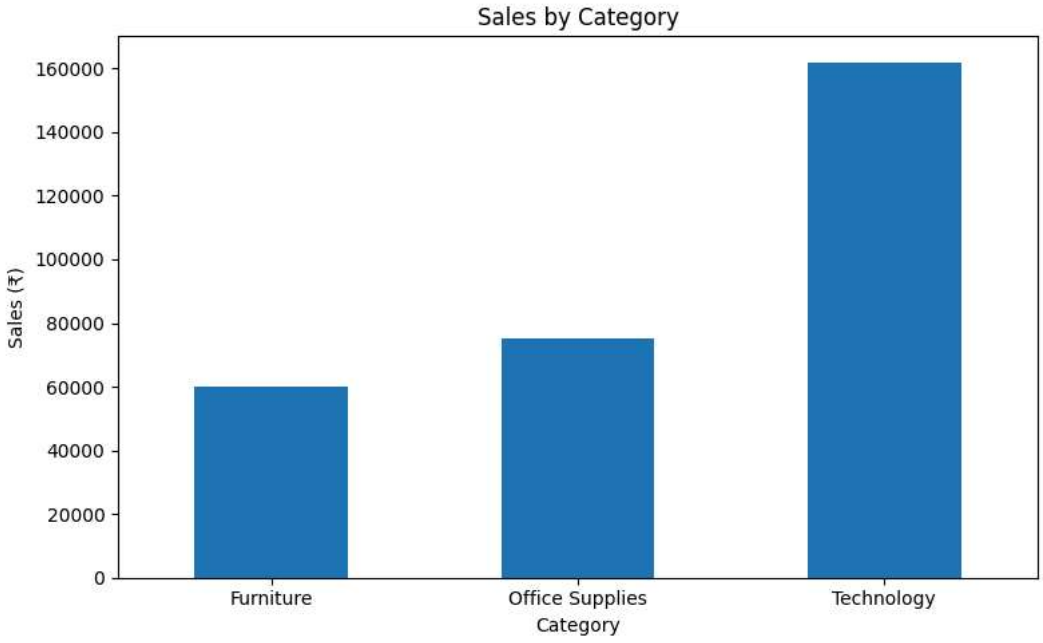
widgets.interact(
    update_dashboard,
    category=category_filter,
    region=region_filter
);
```

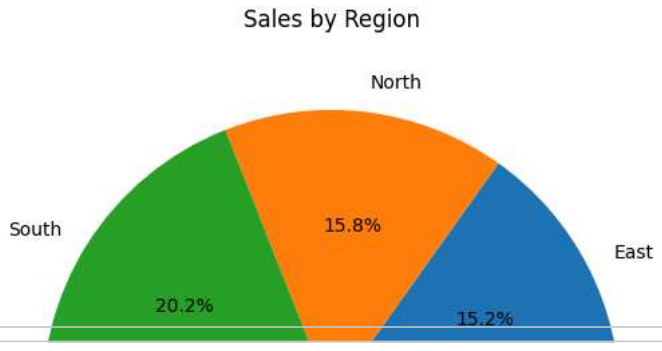
Category: All

Region: All

```
============================================================
             SALES ANALYSIS DASHBOARD
============================================================
Category : All
Region   : All

💰 Total Sales         : ₹ 297,000
📈 Total Profit        : ₹ 53,500
📦 Total Orders        : 10
🛒 Average Order Value : ₹ 29,700
============================================================
```






```
from os import PRIO_PGRP
print("==== SALES ANALTSIS SUMMARY =====")
```

```
print("Total_Sales  :","$",total_sales)
print("Total_Profit :","$",total_profit)
print("Total_Orders :","$",total_orders)
print("Total_Avg_Orders_Values :","$",avg_order_value)
```

```
==== SALES ANALTSIS SUMMARY =====
Total_Sales  : $ 297000
```